Overview of Core Services

Explore the core AWS services including compute, storage, databases, networking, and security. Understand fault isolation boundaries to design resilient cloud architectures and prepare for the AWS Developer Associate exam.

One of Amazon Web Services' (AWS) defining characteristics is the breadth of its services, which span a wide range of categories, including compute, storage, databases, networking, machine learning, analytics, security, IoT, and more.

AWS services#

A non-exhaustive list of some important services, organized in categories, is listed below:

- **Compute services:** AWS provides various compute services, including Amazon EC2 (Elastic Compute Cloud) for scalable virtual servers, AWS Lambda for serverless computing, and Amazon ECS (Elastic Container Service) for container orchestration.
- **Storage services:** AWS offers scalable and durable storage solutions such as Amazon S3 (Simple Storage Service) for object storage, Amazon EBS (Elastic Block Store) for block storage, and Amazon EFS (Elastic File System) for file storage.
- **Database services:** AWS provides managed database services, such as Amazon RDS (relational database service) for relational databases an Amazon DynamoDB for NoSQL databases.

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- **Networking services:** AWS offers networking services such as Amazon VPC (Virtual Private Cloud) for isolated cloud environments and Amazon Route 53 for domain name system (DNS) services.
- **Machine learning and AI services:** AWS provides a wide range of machine learning and artificial intelligence services, including Amazon SageMaker for building, training, and deploying ML models, Amazon Rekognition for image and video analysis, Amazon Comprehend for natural language processing, and Amazon Polly for text-to-speech conversion.
- **Security services:** AWS offers various security services and features to protect data and applications in the cloud. This includes services like AWS Identity and Access Management (IAM), AWS Key Management Service (KMS), and AWS WAF (Web Application Firewall).
- **Management and governance services:** AWS provides management and governance services to help customers manage their AWS resources and environments effectively. This includes services like AWS CloudFormation for infrastructure as code, AWS Config for resource inventory and configuration management, and AWS CloudWatch for monitoring and observability.

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AWS core services

Apart from these core services, AWS offers a wide variety of other services. To make resilient architectures using these services, AWS offers multiple fault isolation constructs. One such construct is the categorization of AWS services among divisions, which helps to predict the scope of impact of the failure of a service.

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Fault isolation boundary

AWS maintains a fault isolation boundary for its services. A **fault isolation boundary** defines the scope within which the fault is contained. The AWS services are divided into three main categories based on their fault isolation boundary:

- **Zonal services:** The resources of zonal services lie within a single availability zone. The resources in one availability zone exist and fail independently of the resources in the other zone. An example of zonal services would be EC2 instances.
- **Regional services:** The resources of regional services span across multiple availability zones and form a single access point for that region. An example of regional services would be the AWS DynamoDB table.
- **Global services:** The resources of global services are not region-specific. An example of a global service would be IAM.

The illustration given below will help to better understand the fault isolation boundary.

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